Nowcast Data Sources — Mool

> All sources relevant to the parametric trigger nowcast. Backtest uses the same sources where historical depth exists.

Rainfall & Water Deficit

Source	Variable	Resolution	Cadence	Lag	Access	Nowcast Role
CHIRPS	Daily precipitation	5 km	Daily	~5 days	GEE, free	Rolling 30/60-day rainfall deficit vs. historical normal
GPM IMERG Early	Near-real-time precipitation	10 km	30 min	~6 hours	GEE, free	Same-day rainfall signal; faster than CHIRPS
ERA5-Land	Precip, evaporation	9 km	Daily	~5 days	GEE, free	Backup rainfall + actual evapotranspiration

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Crop Health & Vegetation

Source	Variable	Resolution	Cadence	Lag	Access	Nowcast Role
Sentinel-2 SR	NDVI, EVI, NDRE, SWIR indices	10–20 m	~5 days	1–2 days	GEE, free	Primary optical crop stress signal; cloud-limited during monsoon
AWIFS_BOA (Bhoonidhi)	Green, Red, NIR, SWIR → NDVI	56 m	~5 days when cloud-free	1–2 days	Bhoonidhi API, whitelisted IP	**PMFBY-aligned sensor** — trigger calibrated on same data as government adjudication
AWIFS 15-day NDVI composite (Bhoonidhi)	Cloud-minimised NDVI mosaic	100 m	15 days	~2 days	Bhoonidhi API, whitelisted IP	**Best cloud-robust NDVI for monsoon season** — directly usable as trigger input
LISS3_L2 (Bhoonidhi)	Green, Red, NIR,	23.5 m	~24-day repeat; 100+	1–2 days	Bhoonidhi API,	Fine-resolution crop

Source	Variable	Resolution	Cadence	Lag	Access	Nowcast Role
	SWIR		scenes/season over Vidarbha		whitelisted IP	discrimination; cross-validates AWIFS
MODIS MOD13Q1	16-day NDVI/EVI composite	250 m	16 days	~2 days	GEE, free	Long historical baseline (2000–present); cloud-free composite; fills S2 gaps
VIIRS VNP13A1	16-day NDVI/EVI	500 m	16 days	~2 days	GEE, free	Higher-frequency backup to MODIS, newer sensor

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Soil Moisture & Heat Stress

Source	Variable	Resolution	Cadence	Lag	Access	Nowcast Role
ERA5-Land	Soil moisture (3 layers), temp, VPD, solar radiation	9 km	Daily	~5 days	GEE, free	Core heat/drought stress variables; VPD is the best single crop-stress scalar
SMAP L3	Surface + root-zone soil moisture	36 km	Daily	~3 days	GEE, free	Independent soil moisture validation; coarse but direct microwave measurement
MODIS LST (MOD11A2)	Land surface temperature	1 km	8 days	~2 days	GEE, free	Heat stress proxy; catches localised temperature anomalies ERA5 misses at 9 km

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SAR / All-Weather

Source	Variable	Resolution	Cadence	Lag	Access	Nowcast Role
Sentinel-1 GRD	VV, VH backscatter	10 m	6–12 days	1 day	GEE, free	Cloud-penetrating crop structure + soil moisture proxy; critical

Source	Variable	Resolution	Cadence	Lag	Access	Nowcast Role
						during monsoon when optical fails
NISAR SSAR GUNW (Bhoonidhi)	InSAR ground displacement	5.6 m	12-day repeat	~3 days	Bhoonidhi API, whitelisted IP	Subsurface irrigation stress, crop height change, waterlogging detection — data starts July 2026

Topography (Static)

Source	Variable	Resolution	Access	Nowcast Role
CartoSat-1 DEM (Bhoonidhi)	Elevation, slope, aspect	30 m	Bhoonidhi API, whitelisted IP	Permanent drainage/waterlogging risk modifier for plot-level scoring; 18 tiles cover all of Vidarbha, all Online=Y
SRTM DEM	Elevation, slope	30 m	GEE, free	Fallback if CartoSat unavailable; slightly older but identical use

Trigger Logic

The arithmetic nowcast stacks sources in three layers:

1. Stress signal — updated every 15 days AWIFS 15-day NDVI composite or Sentinel-2 NDVI → compute anomaly against MODIS/S2 historical median for that calendar week.

2. Forcing signal — updated every few days GPM IMERG rolling 30-day rainfall deficit + ERA5 VPD current value → determines whether stress is drought-driven, heat-driven, or both.

3. Permanent risk modifier — static CartoSat DEM slope → adjusts trigger threshold per plot based on drainage characteristics.

Trigger condition: `NDVI anomaly ≤ −X%` AND `rainfall deficit ≥ Y mm` AND `VPD ≥ Z kPa` → payout triggered.

Access Notes

Access type	What it means
GEE, free	Google Earth Engine — works from any IP, just needs a GEE-authorised Google account

Access type	What it means
Bhoonidhi API, whitelisted IP	ISRO Bhoonidhi portal — requires static public IPv4 to be registered with NRSC at bhoonidhi@nrsc.gov.in

To add a new IP, email: `bhoonidhi[at]nrsc[dot]gov[dot]in` with subject "Bhoonidhi API access - [username], [IP], /auth/token".